Theory of Computation

Pushdown Automata

A PDA is a 7-tuple (Q, Σ, Γ, δ, q0, Z0, F). It is an ε-NFA with a stack: each move reads an input symbol or ε, pops the top stack symbol, and pushes a string of stack symbols.
Acceptance can be by final state or by empty stack; the two are equivalent for nondeterministic PDAs.

PDA for { w c w^R }

Push every symbol of w, switch state on c, then pop and match each symbol of w^R. Accept when the stack is back to Z0.

Equivalence with CFGs

For every CFG there is a PDA accepting the same language (simulate leftmost derivations on the stack), and every PDA language is context-free. Deterministic PDAs accept a strict subset: the DCFLs.